

Multiple Choice

1. **Answer:** A

Options: A) 5,000 pF; B) 7,000 pF; C) 9,500 pF; D) 8,000 pF; E) 5,500 pF

Note: Correct option: A - 5,000 pF

2. **Answer:** B

Options: A) 465 cfs; B) 476 cfs; C) 490 cfs; D) 430 cfs; E) 450 cfs

Note: Correct option: B - 476 cfs

3. **Answer:** A

Options: A) $2 bE(X)$; B) $E(X^2) - [E(X)]^2$; C) $2VarX + b$; D) $VarX - b^2$; E) $VarX + 2 bVarX$

Note: Correct option: A - $2 bE(X)$

4. **Answer:** A

Options: A) $h(y) = [2 / \{\pi(1 + \sqrt{y})\}]$ for $y > 0$ and $= 0$ otherwise; B) $h(y) = [2 / \{\pi(1 + y)\}]$ for $y > 0$ and $= 0$ otherwise; C) $h(y) = [1 / \{\pi(1 + y^2)\}]$ for $y > 0$ and $= 0$ otherwise; D) $h(y) = [1 / \{2\pi(1 + \sqrt{y})\}]$ for $y > 0$ and $= 0$ otherwise; E) $h(y) = [1 / \{2\pi(1 + y^2)\}]$ for $y > 0$ and $= 0$ otherwise

Note: Correct option: A - $h(y) = [2 / \{\pi(1 + \sqrt{y})\}]$ for $y > 0$ and $= 0$ otherwise

5. **Answer:** A

Options: A) 24.2×10^3 psi; B) 27.9×10^3 psi; C) 18.5×10^3 psi; D) 16.3×10^3 psi; E) 22.8×10^3 psi

Note: Correct option: A - 24.2×10^3 psi

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(a) 175 volts, (b) 3,500 amp, (c) 1,750 amp'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'doubled'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(3/4)'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '14.5 GHz'.

Long Form Response

11. **Answer:** $k^2 \geq [\{\pi\} / \{f_1(2 f_m - 1)\}]$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 7. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key